Soroush H. Zargarbashi

Hamburg, Germany soroushzargar@gmail.com +491777447962 soroushzargar.figma.site
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About Me

Researcher in Trustworthy AI, with a focus on uncertainty quantification, conformal prediction, and adversarial robustness. Published in leading ML venues including ICML, ICLR, NeurIPS, and UAI, with industry experience spanning LLM fine-tuning, distributed machine learning, speech AI, and time-series forecasting.

Experiences

Microsoft, Applied Scientist Intern — Expected 2026

- Incoming Applied Scientist Intern, pending visa approval

Apple, Research Intern - Speech Translation Team — Jun - Aug 2025

- Fine-tuned large language models (LLMs) for real-time speech translation
- Built distributed multi-GPU training pipelines using data and model sharding techniques.
- Developed speech embedding representations and data generation pipelines to support LLM fine-tuning.

CISPA Helmholtz Center for Information Security, AI Researcher — Sep 2022 – Present

- Leading research on Trustworthy AI, developing novel methods for uncertainty quantification, conformal prediction, and adversarial robustness.
- Published in top-tier venues: ICML, ICLR, NeurIPS, and UAI.

Dorfak Intelligent Trading - Quantitative Research, Senior Backend Developer — 2021 – 2022

- Designed and implemented microservice-based backend systems for scalable, intelligent trading platforms, covering data curation and strategy management.
- Built AI-powered algorithmic trading strategies for financial markets.
- Utilized Django REST Framework and Docker to build and deploy containerized, production-ready services.

Publications

Should We Pay This Much for Robustness? Efficient Proxy Certificates with Marginal Guarantees | *Under review - NeurIPS 2026* — 2026

Soroush H. Zargarbashi[=], Mohammad Sadegh Akhondzadeh[=], *Aleksandar Bojchevski*

Probably Approximately Guaranteed Algorithms via Graph Neural Networks | *Under review - LoG 2026* — 2026

Soroush H. Zargarbashi, *Aleksandar Bojchevski*, *Christopher Morris*

Conformal Adaptive Test-time Scaling | *Under review - NeurIPS 2026 / Best poster ICLR 2026 workshop* — 2026

Mohammad Sadegh Akhondzadeh, Soroush H. Zargarbashi, Simone Antonelli, *Aleksandar Bojchevski*,

One Sample is Enough to Make Conformal Prediction Robust [Paper] — NeurIPS 2025

Soroush H. Zargarbashi, Mohammad Sadegh Akhondzadeh, Aleksandar Bojchevski

Quantifying Epistemic Predictive Uncertainty in Conformal Prediction — UAI 2026

Siu Lun Chau, Soroush H. Zargarbashi, Yusuf Sale, *Michele Caprio*,

Robust Conformal Prediction with a Single Binary Certificate [Paper] ICLR 2025
Soroush H. Zargarbashi, Aleksandar Bojchevski

Robust Yet Efficient Conformal Prediction Sets [Paper] ICML 2024
Soroush H. Zargarbashi, Mohammad Sadegh Akhondzadeh, Aleksandar Bojchevski

Conformal Inductive Graph Neural Networks [Paper] ICLR 2024
Soroush H. Zargarbashi, Aleksandar Bojchevski

Conformal Prediction Sets for Graph Neural Networks [Paper] ICML 2023
Soroush H. Zargarbashi, Simone Antonelli, Aleksandar Bojchevski

Front-Loaded Robust Conformal Prediction: Heavy Calibration and Light Weight Inference ICML 2026
Soroush H. Zargarbashi, Mohammad Sadegh Akhondzadeh, *Aleksandar Bojchevski*,

EvA: Evolutionary Attack on Graphs [Paper] ICLR 2026
Sadegh Akhondzadeh[=], *Soroush H. Zargarbashi*[=], *Jimin Cao*, *Aleksandar Bojchevski*

Optimal Conformal Prediction under Epistemic Uncertainty [Paper] 2025
Alireza Javanmardi[=], *Soroush H. Zargarbashi*[=], *Aleksandar Bojchevski*, *Willem Waegeman*, *Eyke Hüllermeier*

A Multi-Modal Feature Embedding Approach to Diagnose Alzheimer Disease from Spoken Language [Paper] 2019
Soroush H. Zargarbashi, Bagher BabaAli

Awards

- **Best Poster Award**, ICLR 2026 Workshop: "Catch, Adapt, and Operate: Monitoring ML Models Under Drift" for **CATS: Conformalized Adaptive Test-time Scaling**

Education

Ph.D. **CISPA Helmholtz Center for Information Security - University of Cologne**, Artificial Intelligence — 2022 – 2026
- Thesis: Distribution-free Uncertainty Quantification for Graphs and Adversarial Data
- Supervisor: Prof. Dr. Aleksandar Bojchevski
- Final grade: **Summa cum laude**

M.Sc. **University of Tehran**, Artificial Intelligence — 2017 – 2020
- Thesis: Speech Enhancement under Adverse Acoustic Conditions
- Supervisors: Dr. Bagher BabaAli and Prof. Dr. Yu Tsao (Academia Sinica)

B.Sc. **Isfahan University of Technology**, Computer Engineering — 2012 – 2017
- Thesis: Normalized Dot Product Representation of Graphs

Academic Services

Academic Reviewing and Committee Work

- Reviewer: *NeurIPS (2024, 2025)*, *ICLR (2025)*, *ICML (2025)*
- Program Committee member: *Epistemic Intelligence in Machine Learning Workshop at EurISP 2025*.

Invited Talks & Selected Presentations

- **Log Meetup - Aachen:** Conformal Prediction for Graphs

- **Algorithmic Earth System Science Workshop:** Conformal Prediction in Noisy Conditions
- **SPP2298 Annual meeting:** Conformal Prediction Graph-aware and Robust
- **MLBoost Reading Group:** Robust Yet Efficient Conformal Prediction Sets
- **MLSec Reading Group:** On Robust Conformal Prediction: Single Certificate and Sample

Teaching Assistant

- University of Cologne: *Advanced Machine Learning*, *Elements of Machine Learning*
- Isfahan Univ of Tech: *Algorithm Design*, and *Data Structures*, *Advanced Programming (lab instructor)*

Skills

Machine Learning & AI: Distribution-free Uncertainty Quantification, Conformal Prediction (Split/Full CP, Conformal Risk Control), Adversarial Robustness & Certification (Randomized Smoothing), Epistemic Uncertainty (Credal Sets), Graph Neural Networks (GNNs), Large Language Models (LLMs), Multimodal Learning (Audio–Text), Bayesian Inference, Ensemble Methods, Test-Time Adaptation & Scaling.

Research & Statistical Methods: Distribution-free & PAC-style Theoretical Guarantees, Exchangeability Theory, Adversarial Attack Design (Black-box, Evolutionary Optimization), Statistical Calibration, Active Learning, Selective Classification.

Deep Learning Frameworks: PyTorch (Lightning, DistributedDataParallel), Hugging Face Transformers, PyTorch Geometric, TensorFlow, scikit-learn

Speech & Audio Processing: Speech Embeddings (i-vector, x-vector), Multi-modal Feature Fusion, Acoustic & Syntactic Analysis, Speech-to-Speech Translation, LLM Fine-tuning for Speech.

Programming & Systems: Python, C++, Bash, Docker, REST APIs, Git

References

Prof. Dr. Aleksandar Bojchevski — Professor, University of Cologne (Trustworthy AI)
Email: a.bojchevski[at]uni-koeln[dot]de | Web: abojchevski.github.io

Dr. Matthias Sperber — Apple, Machine Translation. Email: m_sperber[at]apple[dot]com